

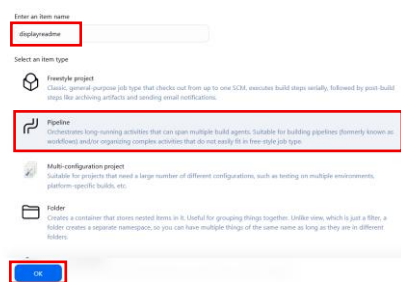
To Begin with the Lab

Job Setup for CloudFormation Parameters in Jenkinsfile Summary of the Lab

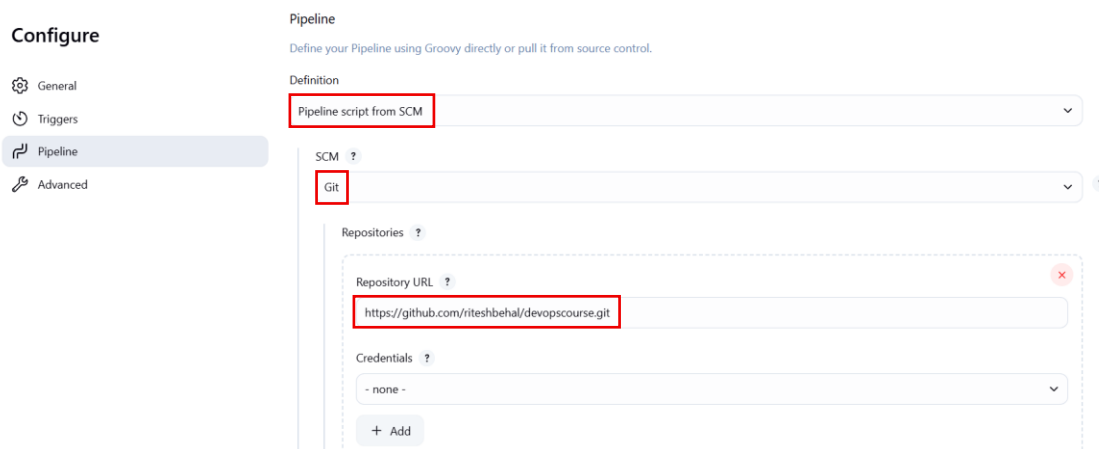
In this lab, we use files from the [CloudFormation](#) GitHub repository to deploy infrastructure using Jenkins and AWS CloudFormation. A new Jenkins pipeline job is created using "Pipeline script from SCM," specifying the repository, branch (main), and correct script path. Parameters like region and file names are passed at build time. The Jenkinsfile executes a CloudFormation deployment. Later, 02_wordpress_Jenkinsfile is updated to hardcode parameters (e.g., key pair and region), committed, and rebuilt. AWS credentials and permissions must be configured. The stack creation results in EC2 instance provisioning.

Here we will use the files [01_s3cft.yml](#), [02_ec2cft.yml](#), [ec2_Jenkinsfile](#), and [s3_Jenkinsfile](#) from the root directory of the [CloudFormation](#) repository

- Create a new Item.
- Give the name.
- Select the pipeline option.
- Click OK



- Under the pipeline, in definition, select the option Pipeline script from SCM.
- Under SCM, Select Git.
- Paste the URL of the Github Repository.



- Set the branch to main.
- Give the script path

- Note: give the path properly in the script path.
- If you want to create the s3 bucket then give the path for the s3_Jenkinsfile and for the ec2 give the path for ec2_Jenkinsfile.
- Click Save and click on build now.

Configure

General

Triggers

Pipeline

Advanced

Branches to build ?

Branch Specifier (blank for 'any') ?

/main

Add Branch

Repository browser ?

(Auto)

Additional Behaviours

Add

Script Path ?

☒ Lightweight checkout ?

- Once the job is created, you will get the option 'build with parameter'
- Click again on 'build with parameter'.
- You will get options like this.
- Give the names of the file properly and choose the current region that your instance is running on.
- Click on build.

Status

Changes

Build with Parameters

Configure

Delete Pipeline

Stages

Rename

Pipeline Syntax

Pipeline exampleparameter

This build requires parameters:

STACK_NAME

Enter the CloudFormation Stack Name.

example-stack

TEMPLATE_NAME

Enter the CloudFormation Template File Name

01_s3cft.yml

REGION

AWS Account Region

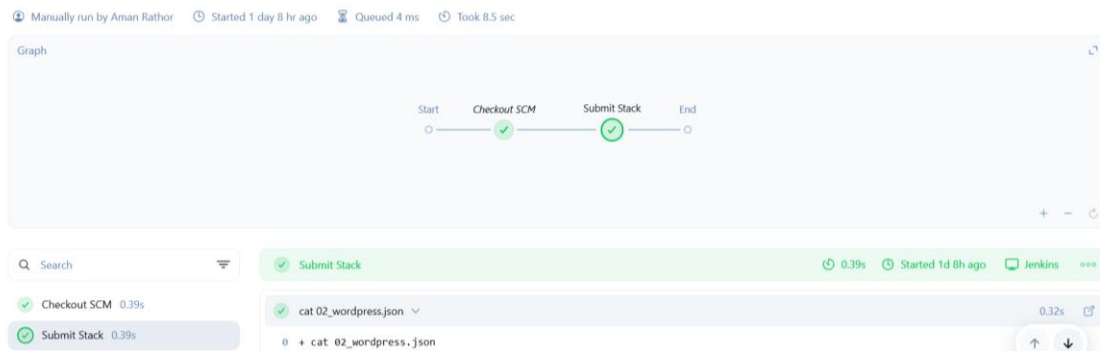
Build Cancel

Builds

Filter

Today

#1 11:07 AM



- Here change the template name and choose the region you are in
- Click build.
- Note that you should configure the aws credential on the machine before building this job and also make sure that you should have the permission of CloudFormation.

Permissions Trust relationships Tags Last Accessed Revoke sessions

Permissions policies (6) [Info](#)

You can attach up to 10 managed policies.

Filter by Type

Search

All types

< 1 > ⚙

☐	Policy name	Type	Attached entities
☐	AdministratorAccess	AWS managed - job function	34
☐	AmazonEC2FullAccess	AWS managed	7
☐	AmazonS3FullAccess	AWS managed	31
☐	AmazonS3ReadOnlyAccess	AWS managed	17
☐	AWSCloudFormationFullAccess	AWS managed	1
☐	AWSLambdaRole	AWS managed	2

STACK_NAME

Enter the CloudFormation Stack Name.

example-stack

PARAMETERS_FILE_NAME

Enter the Parameters File Name (Must contain file extension type *.properties)

example-stack-parameters.properties

TEMPLATE_NAME

Enter the CloudFormation Template File Name

02_wordpress.json

REGION

AWS Account Region

us-east-1

Build Cancel

- It will create a stack under CloudFormation

CloudFormation > Stacks

Stacks (18)

Filter by stack name:

Filter status: **Active** View nested

Stack name	Status	Created time	Description
example-stack	CREATE_IN_PROGRESS	2025-07-18 18:01:53 UTC+0530	AWS CloudFormation Sample Template WordPress_Single_Instance: WordPress is web software you can use to create a beautiful website or blog. This template installs WordPress with a local MySQL database for storage. It demonstrates using the AWS CloudFormation bootstrap scripts to deploy WordPress. **WARNING** This template creates an Amazon EC2 instance. You will be billed for the AWS resources used if you create a stack from this template.
stack77	ROLLBACK_COMPLETE	2025-06-29 17:17:25 UTC+0530	Create S3 Bucket
awsstack29061	UPDATE_COMPLETE	2025-06-29 16:58:49 UTC+0530	Create S3 Bucket & SQS Queue

CloudFormation > Stacks > example-stack

Stacks (18)

Filter by stack name:

Filter status: **Active** View nested

Stacks

- example-stack**
2025-07-18 18:01:53 UTC+0530
CREATE_IN_PROGRESS
- stack77
2025-06-29 17:17:25 UTC+0530
ROLLBACK_COMPLETE
- awsstack29061
2025-06-29 16:58:49 UTC+0530
UPDATE_COMPLETE

example-stack

Stack info | Events | Resources | Outputs | Parameters | Template | Chan

Overview

Stack ID: [arn:aws:cloudformation:us-east-1:878893308172:stack/example-stack/2f684fe0-63d3-11f0-9e43-0affed541e0f](#)

Description: AWS CloudFormation Sample Template
WordPress_Single_Instance: WordPress is web software you can use to create a beautiful website or blog. This template installs WordPress with a local MySQL database for storage. It demonstrates using the AWS CloudFormation bootstrap scripts to deploy WordPress. ****WARNING**** This template creates an Amazon EC2 instance. You will be billed for the AWS resources used if you create a stack from this template.

Status: **CREATE_IN_PROGRESS**

Detailed status: **CONFIGURATION_COMPLETE**

Status reason: Root stack

- And this stack will create an instance.

Instances (3) Info

Find Instance by attribute or tag (case-sensitive):

Connect Instance state Actions Launch instances

All states

	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone	Public I
☐	kubernetes	i-067bbcf0e9bffd802	Running	t3.medium	3/3 checks passed	View alarms +	us-east-1c	ec2-54-
☐	Jenkins	i-0ce929966a84908f7	Running	t3.medium	3/3 checks passed	View alarms +	us-east-1c	ec2-3-8
☐		i-0571c2cc8559c8723	Running	t2.small	Initializing	View alarms +	us-east-1b	ec2-54-